

TABLE I
SIMULATION PARAMETERS.

Parameter	Value
LEO constellation	
Altitude (km)	1200
Orbit plane inclination	87.9°
Time slot	1 s
Maximum transmit power	16.8 W per satellite
Maximum antenna gain	34.33 dBi
Downlink frequency	11.7 GHz
Bandwidth	250 MHz
Antenna pattern	[?] $A \exp(\beta\phi)$
Maximum tilt angle	10°
Approximated antenna gain coefficient, (A)	4.0×10^3
Approximated antenna gain coefficient, (B)	-0.0671
GSO satellite	
Altitude (km)	35785.4
Longitude	30.6
Latitude	0
Downlink frequency	11.7 GHz
GSO ground station	
Longitude	30.6
Latitude	0
Minimum elevation angle	10°
Maximum receive gain	38.1 dBi
Noise temperature	240 K
Antenna pattern	[?]
LEO user ground station	
Maximum receive gain	40.95 dBi
Noise temperature	240 K
Antenna pattern	[?]
User demand	50 Mbps
EPFD	
EPFD limit (dB(W/m ² /1 MHz))	-173.4 [?]
Reference antenna diameter	60 cm

VI. EVALUATION

A. Evaluation Overview

This chapter evaluates the performance of the proposed EPFD-Aware Beam Reassociation for Service Recovery (EABR) framework, focusing on its ability to improve user satisfaction and service continuity under the aggregate EPFD constraint. The evaluation consists of three main parts. First, we observe the aggregate EPFD variation during the critical period near the GS, as well as the number of beams that must be shut down to satisfy the EPFD constraint. This analysis confirms the existence of EPFD violation and beam shutdown demand during the critical period, and further supports the rationale of designing a reassociation-based service recovery framework. Second, we compare different interference mitigation methods in terms of service performance and service recovery capability. Finally, we further analyze the impact of different EPFD limits on user satisfaction to evaluate the stability and applicability of EABR under different interference constraint conditions.

B. Simulation Setup

To evaluate the performance of the proposed EABR framework under the NGSO–GSO coexistence scenario, we design a simulation framework that combines STK and MATLAB. This framework is used to construct the LEO constellation, obtain the time-varying satellite geometry, and further evaluate the performance of different interference mitigation methods in terms of EPFD compliance, downlink capacity, and user satisfaction.

The simulation uses the following two main tools:

- **STK (Satellite Tool Kit):** [?] STK is used to construct the OneWeb-like LEO constellation and generate the relative positions, visibility relationships, and beam coverage information among the LEO satellites, the GS, and users at each time slot.
- **MATLAB:** [?] MATLAB is used to process the system information generated by STK and perform offline preprocessing. This includes constructing the beam–user association, critical/helper satellite relationships, time-slot beam role records, and candidate beam relation graphs. MATLAB is then used to execute the proposed EABR framework and calculate the EPFD and user satisfaction performance of each method at different time slots.

The main simulation parameters used in this chapter are summarized in Table ???. The table integrates the settings related to the LEO constellation, GSO satellite, GS, LEO user ground station, and EPFD evaluation. These parameters are used in the following EPFD evaluation and user satisfaction comparison.

C. Compared Schemes

To evaluate the proposed EABR framework, we compare it with several interference mitigation methods. All methods are evaluated under the same simulation parameters to ensure a fair comparison. The compared schemes are described as follows:

- **Beam shutdown only:** When the aggregate EPFD exceeds the limit, the high-interference beams on critical satellites are directly shut down to satisfy the EPFD constraint. This method does not perform beam reassociation for service recovery.
- **PC + Tilt:** This method represents a power-control and satellite-tilting based interference mitigation strategy. It reduces the interference toward the GS by adjusting the transmit power and satellite tilt angle. Although this method can help satisfy the EPFD constraint, it does not provide a reassociation-based service recovery mechanism for users.
- **Only HBR:** This method only applies HBR. Helper recovery beams serve users located under critical closed beams using the default power of the helper satellite. SBR and released power allocation are not applied.
- **EABR (Proposed):** This method includes SBR, released power allocation for helper recovery beams, and HBR.

D. Evaluation Results

Based on the simulation setup and compared schemes described above, this section evaluates the performance of EABR during the critical period. The analysis is divided into four parts. First, the aggregate EPFD and the number of closed beams are examined to verify whether EABR can still satisfy the EPFD constraint after reassociation-based service recovery, and to observe the beam shutdown behavior of critical satellites at different time slots. Second, the average satisfaction of different methods is compared under different user loads to analyze how increasing system load affects service performance. Third, the number of recovered closed-beam users is used to explain the main reason why EABR

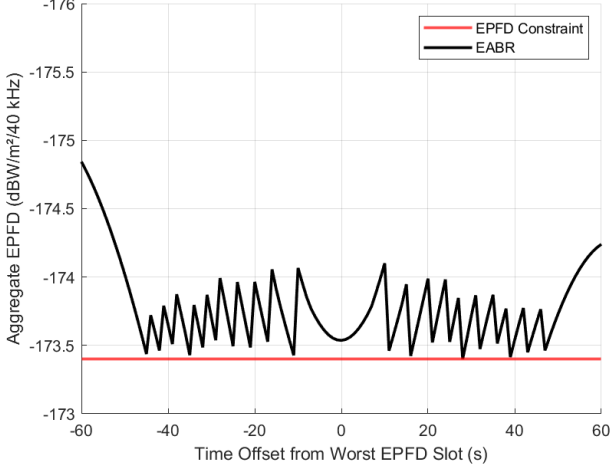


Fig. 1. Aggregate EPFD over time during the critical period.

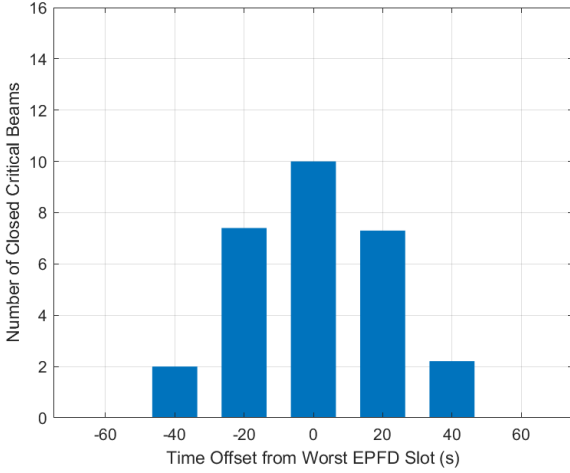


Fig. 2. Number of closed beams on critical satellites over time during the critical period.

improves satisfaction, and the satisfaction CDF is further used to observe the satisfaction distribution of all users. Finally, the average user satisfaction of EABR under different EPFD limits is evaluated to observe how the strictness of the interference limit affects service performance.

1) EPFD Compliance and Beam Shutdown:

This subsection first analyzes the aggregate EPFD after applying EABR and further observes the number of beams that must be shut down on critical satellites to satisfy the EPFD constraint. This analysis is used to verify whether the proposed framework can still satisfy the EPFD limit after reassociation-based service recovery, and to show the beam shutdown behavior during the critical period.

Fig. ?? shows the aggregate EPFD variation during the critical period near the GS. The red line represents the EPFD constraint, and the black line represents the aggregate EPFD after applying EABR. As shown in the figure, EABR keeps the aggregate EPFD below the limit during the entire observation period. This indicates that the proposed framework does not

cause EPFD violation while performing service recovery for closed-beam users. Therefore, EABR can compensate for the service loss caused by beam shutdown while maintaining EPFD compliance.

Fig. ?? shows the number of beams that must be shut down on critical satellites at different time offsets. When critical satellites approach the worst EPFD slot, the number of closed beams gradually increases and reaches its maximum around the worst EPFD slot. After the satellites move away from the GS, the number of closed beams gradually decreases. This result shows that the beam shutdown demand changes with the satellite-GS geometry.

2) User Satisfaction under Different User Loads:

This subsection compares the average satisfaction of different interference mitigation methods under different user loads. By changing the number of users, we can observe how the service performance changes as the system load increases, and evaluate whether EABR can maintain user satisfaction under different load conditions.

Fig. ?? shows the average satisfaction under three user load scenarios: 30 users, 50 users, and 70 users. As shown in the figure, Beam shutdown only has the lowest satisfaction because it does not provide service recovery for users affected by closed beams. PC + Tilt can reduce the interference toward the GS, but it does not recover the service of users affected by beam shutdown. Only HBR can serve part of the closed-beam users through helper recovery beams, so it achieves better satisfaction than Beam shutdown only and PC + Tilt.

Compared with the other methods, EABR maintains higher average satisfaction under all three user loads. When the user load is low, the helper satellite usually has enough available power, so the difference between Only HBR and EABR is small. However, as the user load increases, more closed-beam users need service recovery, and Only HBR becomes limited by the default power of the helper satellite. In contrast, EABR uses SBR to release available power from the helper satellite and improve the service recovery capability of helper recovery beams. Therefore, EABR can maintain more stable service performance under medium and high user loads.

3) Service Recovery Capability and Satisfaction Distribution:

This subsection further analyzes why EABR improves user satisfaction and observes the satisfaction distribution of different methods. Since 30 users is a low-load scenario, the default available power of the helper satellite is usually sufficient to support most recovery demand. Therefore, the difference in service recovery capability is not obvious under 30 users. To better show the effect of the proposed framework, this subsection focuses on the 50-user and 70-user scenarios.

Fig. ?? compares the number of successfully recovered closed-beam users between Only HBR and EABR under the 50-user and 70-user scenarios. As shown in the figure, EABR serves more closed-beam users in both load scenarios. The difference becomes more obvious around the worst EPFD slot, where the beam shutdown demand is the most severe. This result shows the necessity of SBR in EABR.

Fig. ?? shows the satisfaction CDF of three interference mitigation methods. This figure focuses on the user-level

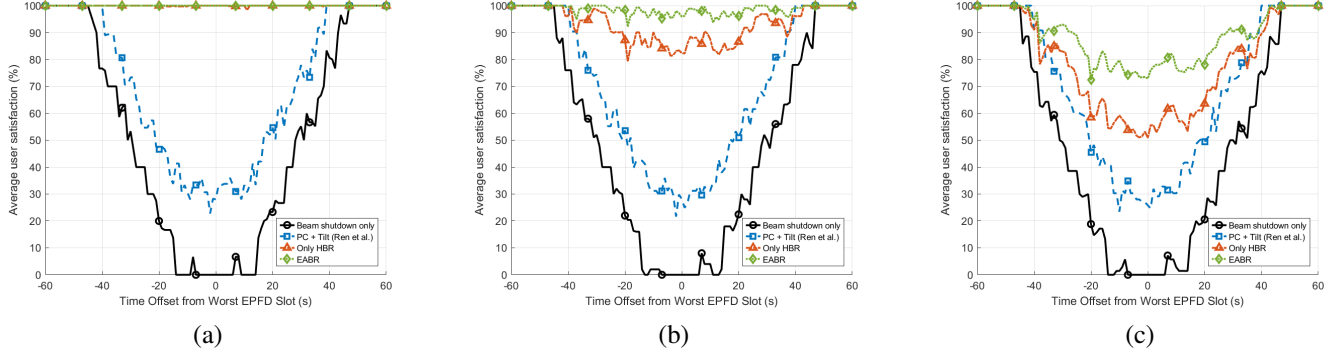


Fig. 3. Average user satisfaction of different interference mitigation methods under different user loads. (a) 30 users. (b) 50 users. (c) 70 users.

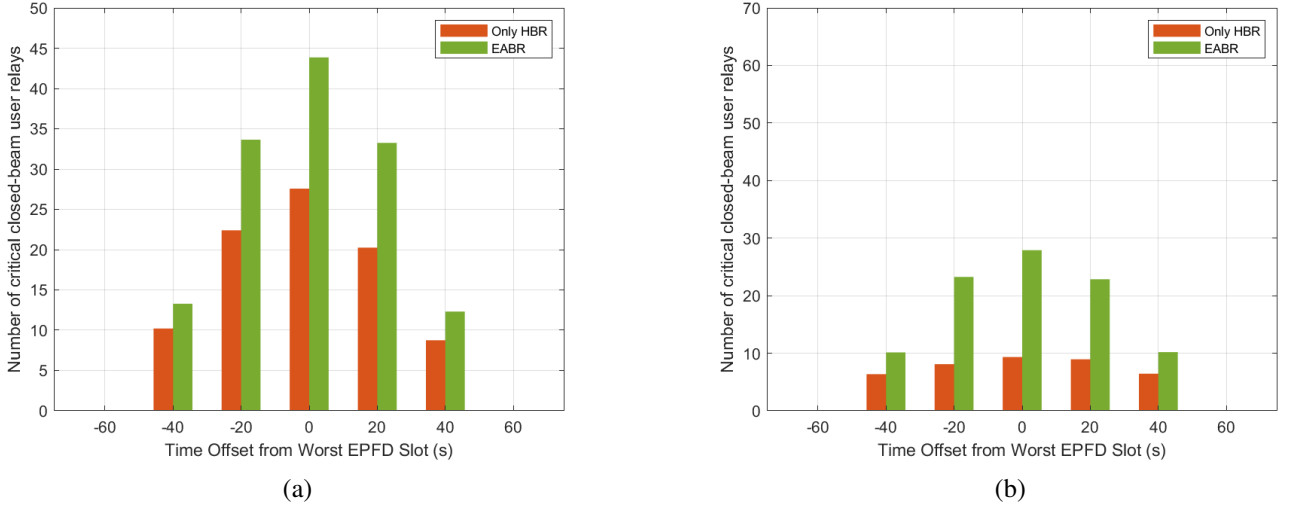


Fig. 4. Number of recovered closed-beam users with and without SBR under different user loads. (a) 50 users. (b) 70 users.

satisfaction distribution. A CDF curve that shifts further to the right means that more users maintain higher satisfaction. In contrast, if the curve rises quickly in the low-satisfaction region, more users suffer from service degradation.

As shown in the figure, the CDF curve of EABR shifts more toward the high-satisfaction region than the other methods. This means that more users can maintain better service quality. The result shows that EABR not only improves average satisfaction, but also reduces the proportion of low-satisfaction users, making the overall user service quality more stable.

4) Impact of Different EPFD Limits:

This subsection further evaluates the impact of different EPFD limits on the service performance of EABR. When the EPFD limit becomes stricter, the allowed aggregate interference becomes lower. As a result, critical satellites may need to shut down more beams to satisfy the constraint, and the recovery demand of closed-beam users may also increase. This analysis is used to observe whether EABR can maintain stable average user satisfaction under different EPFD limits.

Fig. ?? shows the average user satisfaction of EABR under different EPFD limits. As shown in the figure, when the EPFD limit is more relaxed, EABR can maintain higher average user satisfaction. This result shows that the strictness of the EPFD

limit directly affects user satisfaction. It also indicates that EABR can achieve better service performance under more relaxed interference constraints.

Overall, EABR keeps the aggregate EPFD below the limit during the critical period and uses SBR to improve the ability of helper recovery beams to serve closed-beam users. Compared with other interference mitigation methods, EABR maintains higher average satisfaction under different user loads and reduces the proportion of low-satisfaction users. In addition, the results under different EPFD limits show that the strictness of the interference constraint affects user satisfaction, and EABR can maintain better service performance under more relaxed constraints. Overall, EABR achieves a better balance between EPFD compliance and service continuity.

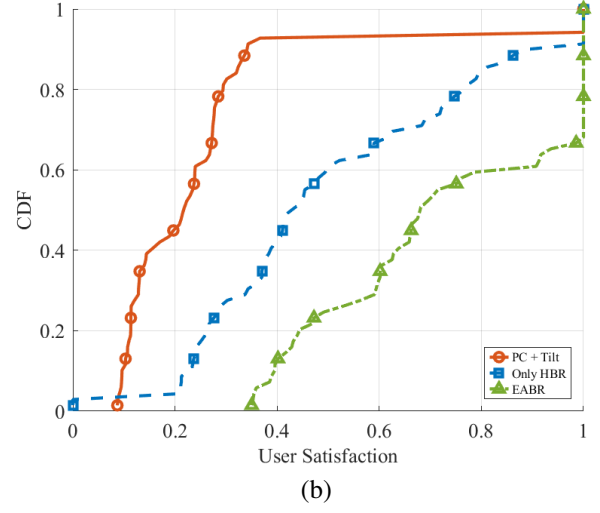
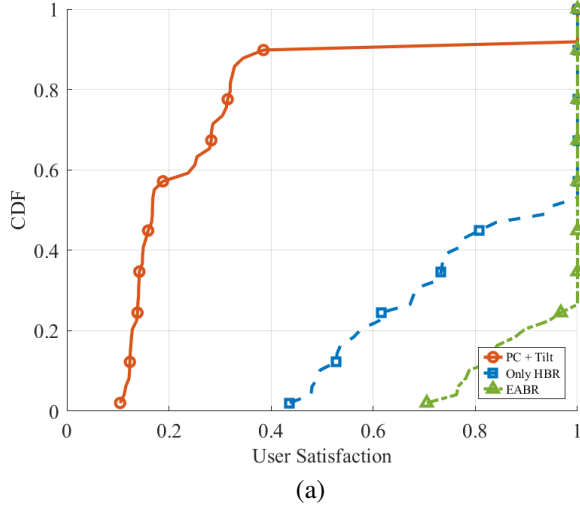


Fig. 5. CDF of user satisfaction for different interference mitigation methods under different user loads. (a) 50 users. (b) 70 users.

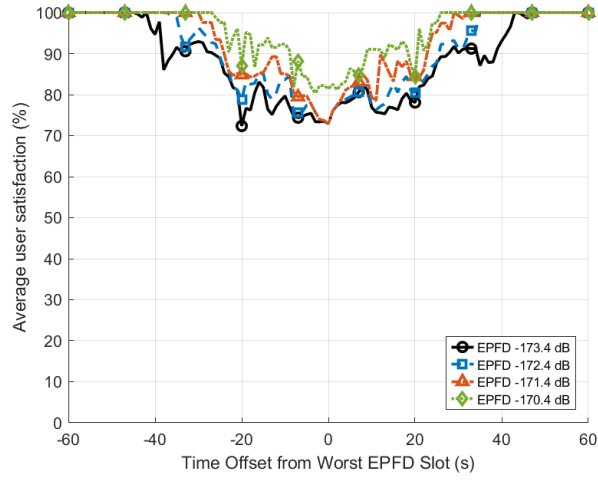


Fig. 6. Average user satisfaction of EABR under different EPFD limits.